

Magnus Victor Boock

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Summary

- PhD student in Computer Science at the University of Southern Denmark, researching offline distributional reinforcement learning with PAC-Bayesian methods since August 2025.

Education

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| PhD | University of Southern Denmark , Computer Science | Odense, Denmark
Aug 2025 – July 2028 |
| | <ul style="list-style-type: none"> • Supervised by Prof. Melih Kandemir. • Preprint: Hitting Time Isomorphism for Multi-Stage Planning with Foundation Policies (arXiv:2605.06470) — a new theoretical framework and algorithm for long-horizon offline RL planning, proposing an operator-theoretic framework that learns asymmetric, triangle-inequality-satisfying state representations from hitting-time observations, used to develop Isomorphic Embedding Learning (IEL), a foundation policy algorithm achieving state-of-the-art results on long-horizon navigation benchmarks via directed graph-based planning. • RLSS 2026: Selected attendee, Reinforcement Learning Summer School, Milan — competitive admission, lectures from leading RL researchers. | |
| MS | University of Southern Denmark , Computer Science | Odense, Denmark
Sept 2023 – June 2025 |
| | <ul style="list-style-type: none"> • Generative Flow Networks for realistic tissue data generation. Supervised by Prof. Richard Röttger and Prof. Melih Kandemir. • Weighted average Danish grade: 10.67/12, GPA converted: 3.76/4. Thesis received the top grade. | |
| BS | University of Southern Denmark , Computer Science | Odense, Denmark
Sept 2020 – June 2023 |
| | <ul style="list-style-type: none"> • Supervised by Assoc. Prof. Luís Cruz-Filipe. • Weighted average Danish grade: 10.38/12, GPA converted: 3.59/4. Thesis received the top grade. | |

Experience

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| ForeFlight, Routing Team | Student Software Developer (.NET) | Odense, Denmark
Mar 2024 – Apr 2025
1 year 2 months |
| | <ul style="list-style-type: none"> • Enhanced routing engine with shortest path and constraint algorithms. • Implemented new functionality, bug fixes and ported internal tools. | |
| NIJ-TA3 ADBOU / SDU | Student Software Developer | Odense, Denmark
July 2023 – Dec 2024
1 year 6 months |
| | <p>The project is based on Transition Analysis 3, a collaborative effort between SDU and the University of Pennsylvania, funded by the US National Institute of Justice.</p> <ul style="list-style-type: none"> • Co-developed a cross-platform desktop app for accurate age estimation based on skeletal remains. • The project resulted in a publication in Forensic Science International. | |
| IMADA / SDU | Teaching Assistant | Odense, Denmark
Sept 2021 – present
5 years |
| | <ul style="list-style-type: none"> • Introduction to Computer Science, Algorithms and Data Structures, Networks and Cyber-security, and Reinforcement Learning. | |

Skills

Research Areas: Reinforcement Learning (Offline, Distributional), Deep Learning, Graph-based Planning, Representation Learning

ML Frameworks & Libraries: PyTorch, JAX, NumPy, SciPy, Scikit-learn

RL Environments & Benchmarks: Gymnasium (MuJoCo), D4RL, OGBench

Programming Languages: Python, C++, C, C#

Tools & Platforms: Git, Linux, Conda, LaTeX, Weights & Biases